

NATIONAL UNIVERSITY OF MODERN LANGUAGES

Faculty of Engineering and Computing
Department of Software Engineering



Lab Tasks

OS

Submitted By	: Ijlal Ahmad FL-21565
Semester	: 4th
Date of Submission	: 09 December 2024

Lab 01

a. What is the file system used by Linux?

- The common file systems used by Linux include **ext4** (most widely used), **ext3**, **ext2**, and others like XFS, Btrfs, and ReiserFS.

b. Name two types of boot loaders available.

- Popular boot loaders for Linux are:
 - **GRUB (GNU GRUB)**: Commonly used in most Linux distributions.
 - **LILO (Linux Loader)**: An older boot loader, though less common now.

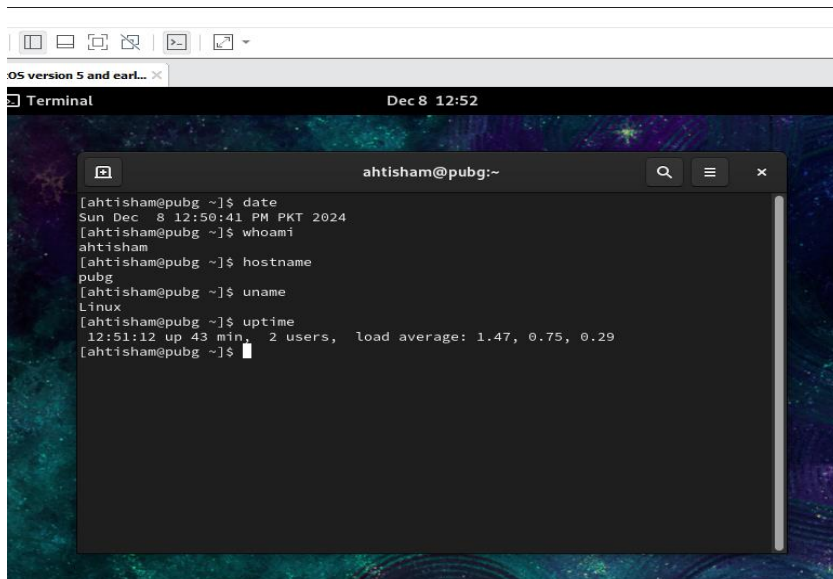
c. What are the names of partitions created for Linux?

- Common partitions created in a Linux system include:
 - **/ (root)**: The main directory containing the Linux operating system.
 - **/home**: Contains user-specific files and settings.
 - **swap**: Used for virtual memory.
 - Optionally, **/boot**: Contains boot-related files, and **/var**, **/tmp**, etc., depending on system configuration.

LAB 02

Q1. Q1 a. Log in. b. Log out. c. Log in again. Open a terminal window, to start a shell. d. Exit from the shell; the terminal window will close. e. Start another shell. Enter each of the following commands in turn.

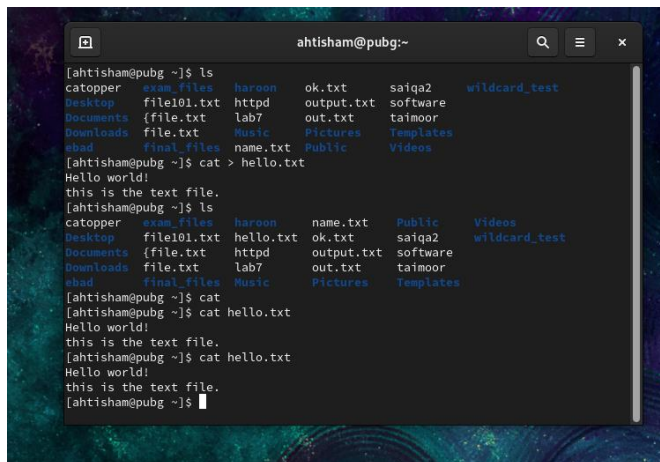
i. date ii. whoami iii. hostname iv. uname v. uptime

A screenshot of a terminal window titled "Terminal" with a timestamp of "Dec 8 12:52". The terminal shows a series of commands and their outputs. The prompt is "ahtisham@pubg:~". The commands and outputs are: "date" returns "Sun Dec 8 12:50:41 PM PKT 2024"; "whoami" returns "ahtisham"; "hostname" returns "pubg"; "uname" returns "Linux"; "uptime" returns "12:51:12 up 43 min, 2 users, load average: 1.47, 0.75, 0.29".

```
[ahtisham@pubg ~]$ date
Sun Dec 8 12:50:41 PM PKT 2024
[ahtisham@pubg ~]$ whoami
ahtisham
[ahtisham@pubg ~]$ hostname
pubg
[ahtisham@pubg ~]$ uname
Linux
[ahtisham@pubg ~]$ uptime
12:51:12 up 43 min, 2 users, load average: 1.47, 0.75, 0.29
[ahtisham@pubg ~]$
```

Q2.

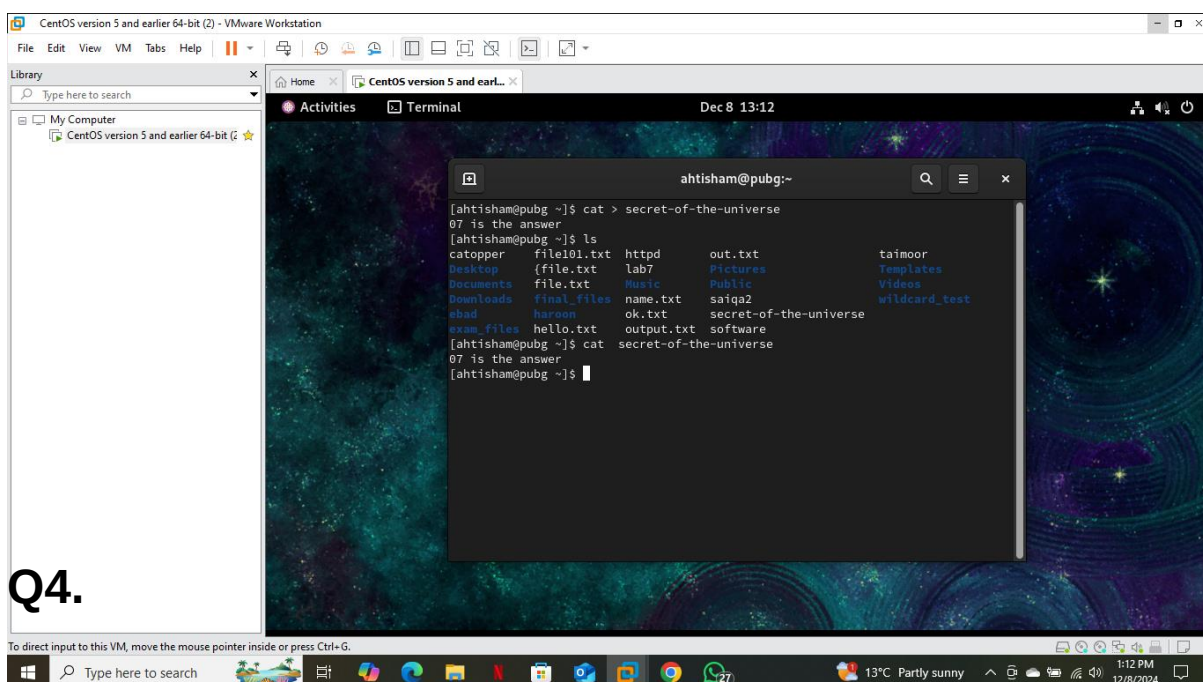
a. Use the ls command to see if you have any files. b. Create a new file using the cat command as follows: \$ cat > hello.txt Hello world! This is a text file. c. Press Enter at the end of the last line, then Ctrl+D to denote the end of the file. d. Use ls again to verify that the new file exists. e. Display the contents of the file. f. Display the file again, but use the cursor keys to execute the same command again without having to retype it.



```
ahtisham@pubg:~$ ls
catoppper  exam_files  haroon  ok.txt  saiqa2  wildcard_test
Desktop    file101.txt httpd    output.txt  software
Documents  {file.txt  lab7    out.txt  taimoor
Downloads  file.txt   Music   Pictures  Templates
ebad       final_files name.txt Public    Videos
[ahtisham@pubg ~]$ cat > hello.txt
Hello world!
this is the text file.
[ahtisham@pubg ~]$ ls
catoppper  exam_files  haroon  name.txt  Public  Videos
Desktop    file101.txt hello.txt ok.txt    saiqa2  wildcard_test
Documents  {file.txt  httpd    output.txt software
Downloads  file.txt   lab7     out.txt   taimoor
ebad       final_files Music     Pictures  Templates
[ahtisham@pubg ~]$ cat
[ahtisham@pubg ~]$ cat hello.txt
Hello world!
this is the text file.
[ahtisham@pubg ~]$ cat hello.txt
Hello world!
this is the text file.
[ahtisham@pubg ~]$
```

Q3.

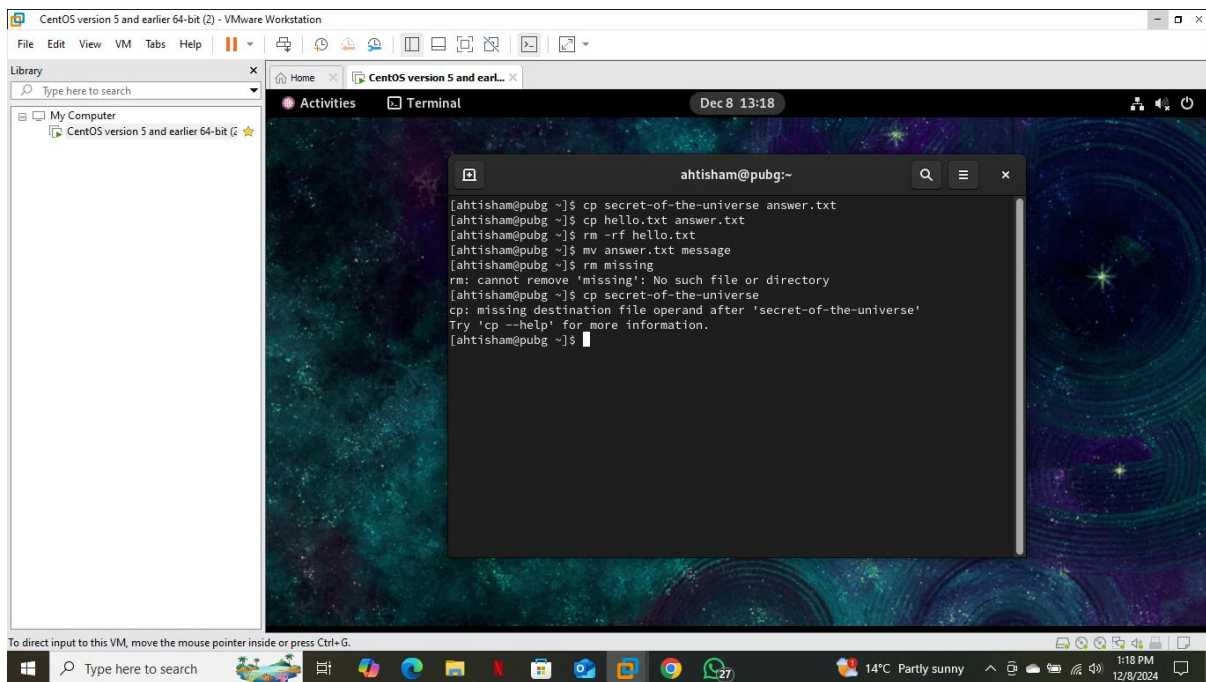
a. Create a second file. Call it secret-of-the-universe, and put in whatever content you deem appropriate. b. Check its creation with ls. c. Display the contents of this file. Minimize the typing needed to do this: i. Scroll back through the command history to the command you used to create the file. ii. Change that command to display secret-of-the-universe instead of creating it.



```
CentOS version 5 and earlier 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
CentOS version 5 and earlier 64-bit (2)
Activities Terminal Dec 8 13:12
[ahtisham@pubg ~]$ cat > secret-of-the-universe
07 is the answer
[ahtisham@pubg ~]$ ls
catoppper  exam_files  haroon  name.txt  Public  Videos
Desktop    file101.txt hello.txt ok.txt    saiqa2  wildcard_test
Documents  {file.txt  httpd    output.txt software
Downloads  file.txt   lab7     out.txt   taimoor
ebad       final_files Music     Pictures  Templates
[ahtisham@pubg ~]$ cat secret-of-the-universe
07 is the answer
[ahtisham@pubg ~]$
```

Q4.

After each of the following steps, use ls and cat to verify what has happened. a. Copy secret-of-the-universe to a new file called answer.txt. Use Tab to avoid typing the existing file's name in full. b. Now copy hello.txt to answer.txt. What's happened now? c. Delete the original file, hello.txt. d. Rename answer.txt to message. e. Try asking rm to delete a file called missing. What happens? f. Try copying secret-of-the-universe again, but don't specify a filename to which to copy. What happens now?

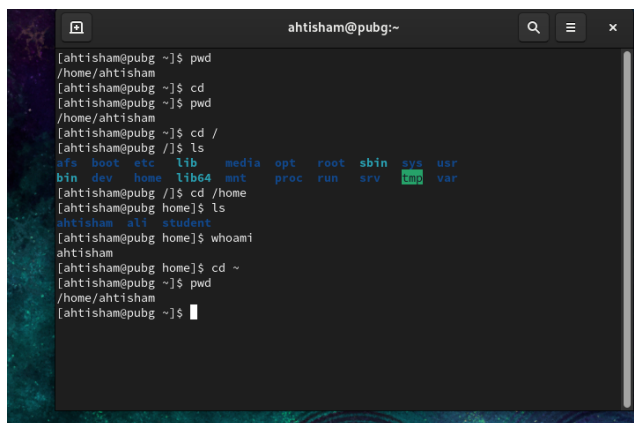


```
CentOS version 5 and earlier 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
CentOS version 5 and earlier 64-bit (2)
Activities Terminal Dec 8 13:18
ahtisham@pubg:~
[ahtisham@pubg ~]$ cp secret-of-the-universe answer.txt
[ahtisham@pubg ~]$ cp hello.txt answer.txt
[ahtisham@pubg ~]$ rm -rf hello.txt
[ahtisham@pubg ~]$ mv answer.txt message
[ahtisham@pubg ~]$ rm missing
rm: cannot remove 'missing': No such file or directory
[ahtisham@pubg ~]$ cp secret-of-the-universe
cp: missing destination file operand after 'secret-of-the-universe'
Try 'cp --help' for more information.
[ahtisham@pubg ~]$
```

LAB 03

Q1.

a. Use the `pwd` command to find out what directory you are in. b. If you are not in your home directory (`/home/USERNAME`) then use `cd` without any arguments to go there, and do `pwd` again. c. Use `cd` to visit the root directory, and list the files there. You should see home among the list. d. Change into the directory called home and again list the files present. There should be one directory for each user, including the user you are logged in as (you can use `whoami` to check that). e. Change into your home directory to confirm that you have gotten back to where you started.

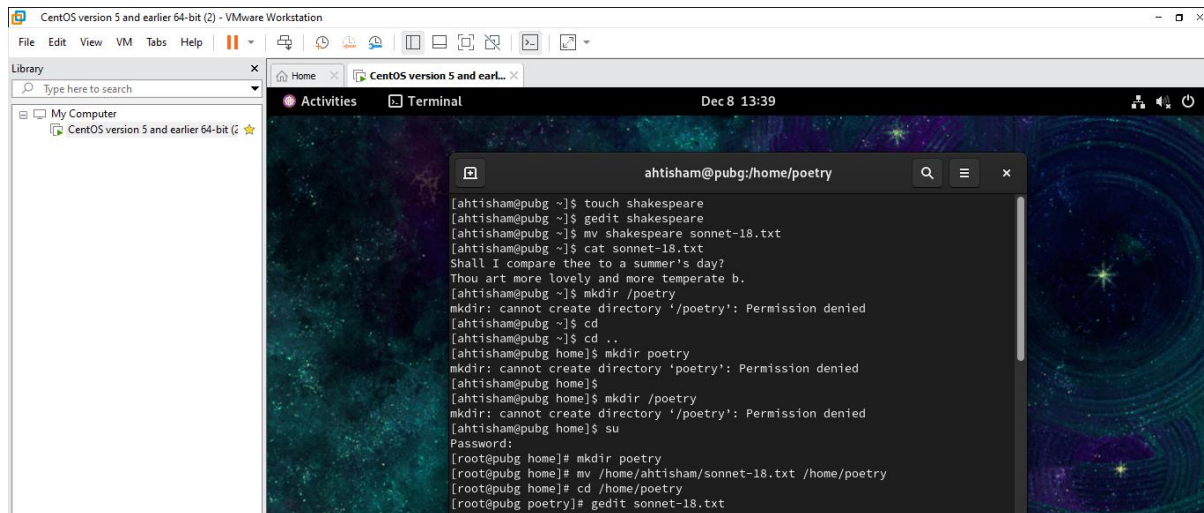


```
ahtisham@pubg:~  
[ahtisham@pubg ~]$ pwd  
/home/ahtisham  
[ahtisham@pubg ~]$ cd  
[ahtisham@pubg ~]$ pwd  
/home/ahtisham  
[ahtisham@pubg ~]$ cd /  
[ahtisham@pubg /]$ ls  
afs  boot  etc  lib  media  opt  root  sbin  sys  usr  
bin  dev  home  lib64  mnt  proc  run  srv  tmp  var  
[ahtisham@pubg /]$ cd /home  
[ahtisham@pubg home]$ ls  
ahtisham  ali  student  
[ahtisham@pubg home]$ whoami  
ahtisham  
[ahtisham@pubg home]$ cd ~  
[ahtisham@pubg ~]$ pwd  
/home/ahtisham  
[ahtisham@pubg ~]$
```

Q2.

a. Create a text file in your home directory called `shakespeare`, containing the following text: Shall I compare thee to a summer's day? Thou art more lovely and more temperate b. Rename it to `sonnet-18.txt`. c. Make a new directory in your home directory, called `poetry`. d. Move the poem file into the new directory. e. Try to find a graphical directory-browsing program, and find your home directory with it. You should also be able to use it to explore some of the system

directories. f. Find a text editor program and use it to display and edit the sonnet.



```
CentOS version 5 and earlier 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
CentOS version 5 and earlier 64-bit (2)
Activities Terminal Dec 8 13:39
ahtisham@pubg/home/poetry
[ahtisham@pubg ~]$ touch shakespeare
[ahtisham@pubg ~]$ gedit shakespeare
[ahtisham@pubg ~]$ mv shakespeare sonnet-18.txt
[ahtisham@pubg ~]$ cat sonnet-18.txt
Shall I compare thee to a summer's day?
Thou art more lovely and more temperate b.
[ahtisham@pubg ~]$ mkdir /poetry
mkdir: cannot create directory '/poetry': Permission denied
[ahtisham@pubg ~]$ cd
[ahtisham@pubg ~]$ cd ..
[ahtisham@pubg home]$ mkdir poetry
mkdir: cannot create directory 'poetry': Permission denied
[ahtisham@pubg home]$ mkdir /poetry
mkdir: cannot create directory '/poetry': Permission denied
[ahtisham@pubg home]$ su
Password:
[root@pubg home]# mkdir poetry
[root@pubg home]# mv /home/ahtisham/sonnet-18.txt /home/poetry
[root@pubg home]# cd /home/poetry
[root@pubg poetry]# gedit sonnet-18.txt
```

Q3.

a. From your home directory, list the files in the directory /usr/share. b. Change to that directory, and use pwd to check that you are in the right place. List the files in the current directory again, and then list the files in the directory called doc. c. Next list the files in the parent directory, and the directory above that. d. Try the following command, and make sure you understand the result: `$ echo ~` e. Use cat to display the contents of a text file which resides in your home directory (create one if you haven't already), using the `~/` syntax to refer to it. It shouldn't matter what your current directory is when you run the command.


```
ahtisham@pubg:/
libgccomp      xorg-x11-xinit
libsecret      xz
libshout       yajl
libsigc++20    yelp
libsigsegv     yelp-tools
libslirp       yelp-xsl
libSM          zenity
libndfile      zip
libnslv        zlib
libsupp        [ahtisham@pubg share]$ cd ..
[ahtisham@pubg usr]$ ls
bin  games  include  lib  lib64  libexec  local  sbin  share  src  tmp
[ahtisham@pubg usr]$ cd ..
[ahtisham@pubg /]$ ls
etc  boot  dev  lib  media  opt  root  sbin  sys  usr
[ahtisham@pubg /]$ echo~
bash: echo~: command not found...
Similar command is: 'echo'
[ahtisham@pubg /]$ echo
[ahtisham@pubg /]$ echo ~
/home/ahtisham
```

Q4.

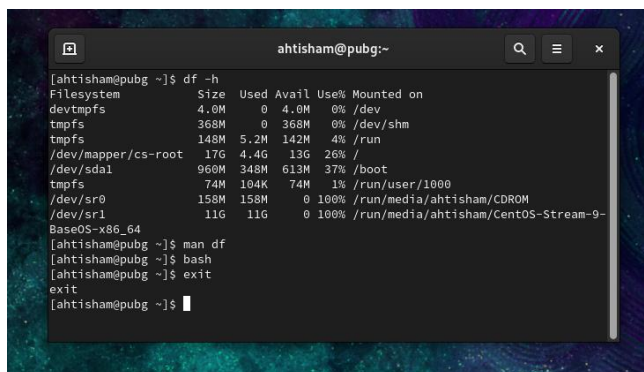
a. Use the hostname command, with no options, to print the hostname of the machine you are using. b. Use man to display some documentation on the hostname command. Find out how to make it print the IP address of the machine instead of the hostname. You will need to scroll down the manpage to the 'Options' section. c. Use the locate command to find files whose name contains the text 'hostname'. Which of the filenames printed contain the actual hostname program itself? Try running it by entering the program's absolute path to check that you really have found it.

```
CentOS version 5 and earlier 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
CentOS version 5 and earlier 64-bit (2)
Activities Terminal Dec 8 17:02
ahtisham@pubg:~
[ahtisham@pubg ~]$ hostname
pubg
[ahtisham@pubg ~]$ man hostname
[ahtisham@pubg ~]$ /usr/bin/hostname
pubg
[ahtisham@pubg ~]$ locate hostname
/etc/hostname
/usr/bin/hostname
/usr/bin/hostnamectl
/usr/bin/nmtui-hostname
/usr/lib/dracut/modules.d/01systemd-hostnamed
/usr/lib/dracut/modules.d/01systemd-hostnamed/99-systemd-networkd-dracut.conf
/usr/lib/dracut/modules.d/01systemd-hostnamed/module-setup.sh
/usr/lib/dracut/modules.d/01systemd-hostnamed/org.freedesktop.hostname1-dracut.conf
/usr/lib/dracut/modules.d/01systemd-hostnamed/systemd-hostname-dracut.conf
/usr/lib/python3.9/site-packages/sos/cleaner/mappings/hostname_map.py
/usr/lib/python3.9/site-packages/sos/cleaner/mappings/__pycache__/hostname_map.cpython-39.opt-1.pyc
/usr/lib/python3.9/site-packages/sos/cleaner/mappings/__pycache__/hostname_map.cpython-39.opt-1.pyc
/usr/lib/python3.9/site-packages/sos/cleaner/parsers/hostname_parser.py
/usr/lib/python3.9/site-packages/sos/cleaner/parsers/__pycache__/hostname_parser.cpython-39.opt-1.pyc
```


LAB 04.

Q1.

- Use the `df` command to display the amount of used and available space on your hard drive.
- Check the man page for `df`, and use it to find an option to the command which will display the free space in a more human-friendly form. Try both the single-letter and long-style options.
- Run the shell, `bash`, and see what happens. Remember that you were already running it to start with. Try leaving the shell you have started with the `exit` command.



```
ahtisham@pubg:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           368M   0  368M   0% /dev/shm
tmpfs           148M  5.2M  142M   4% /run
/dev/mapper/cs-root 17G  4.4G   13G  26% /
/dev/sdal        960M  348M  613M  37% /boot
tmpfs           74M   104K   74M   1% /run/user/1000
/dev/sr0        158M  158M   0 100% /run/media/ahtisham/CDROM
/dev/sr1        11G   11G   0 100% /run/media/ahtisham/CentOS-Stream-9-BaseOS-x86_64

ahtisham@pubg ~]$ man df
ahtisham@pubg ~]$ bash
ahtisham@pubg ~]$ exit
exit
ahtisham@pubg ~]$
```

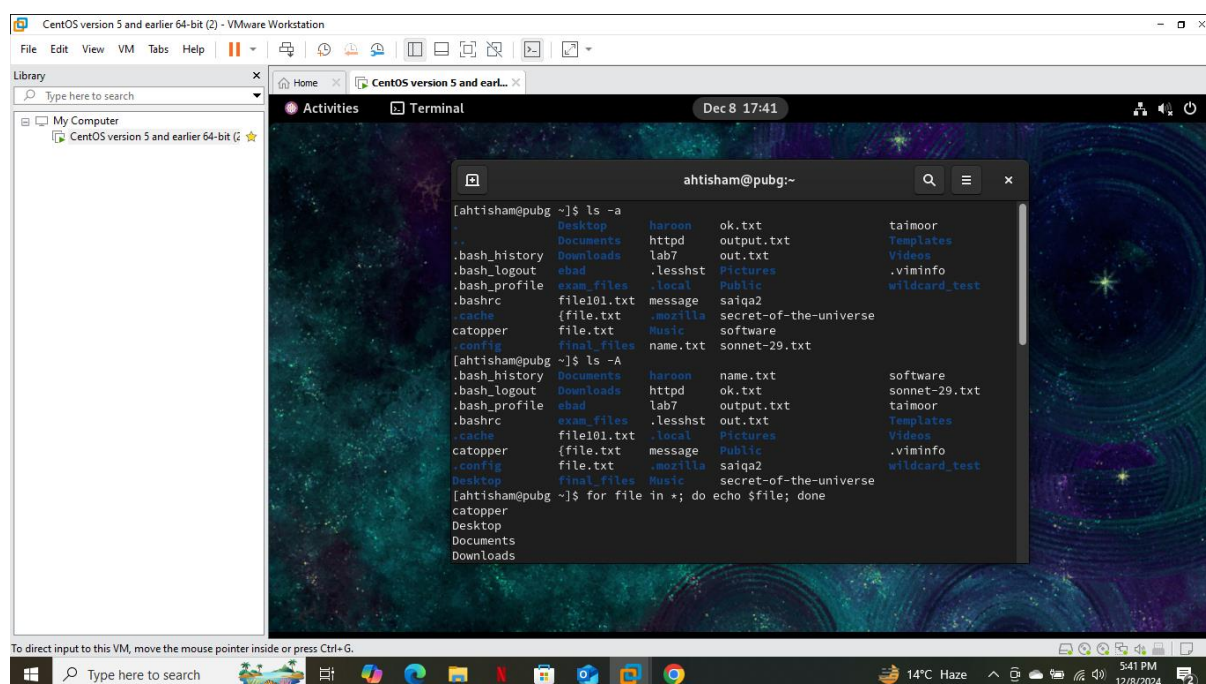
Q2.

a. Try ls with the -a and -A options. What is the difference between them?

b. Write a for loop which goes through all the files in a directory and prints out their names with echo. If you write the whole thing on one line, then it will be easy to repeat it using the command line history.

c. Change the loop so that it goes through the names of the people in the room (which needn't be the names of files) and print greetings to them.

d. Of course, a simpler way to print a list of filenames is echo *. Why might this be useful, when we usually use the ls command?

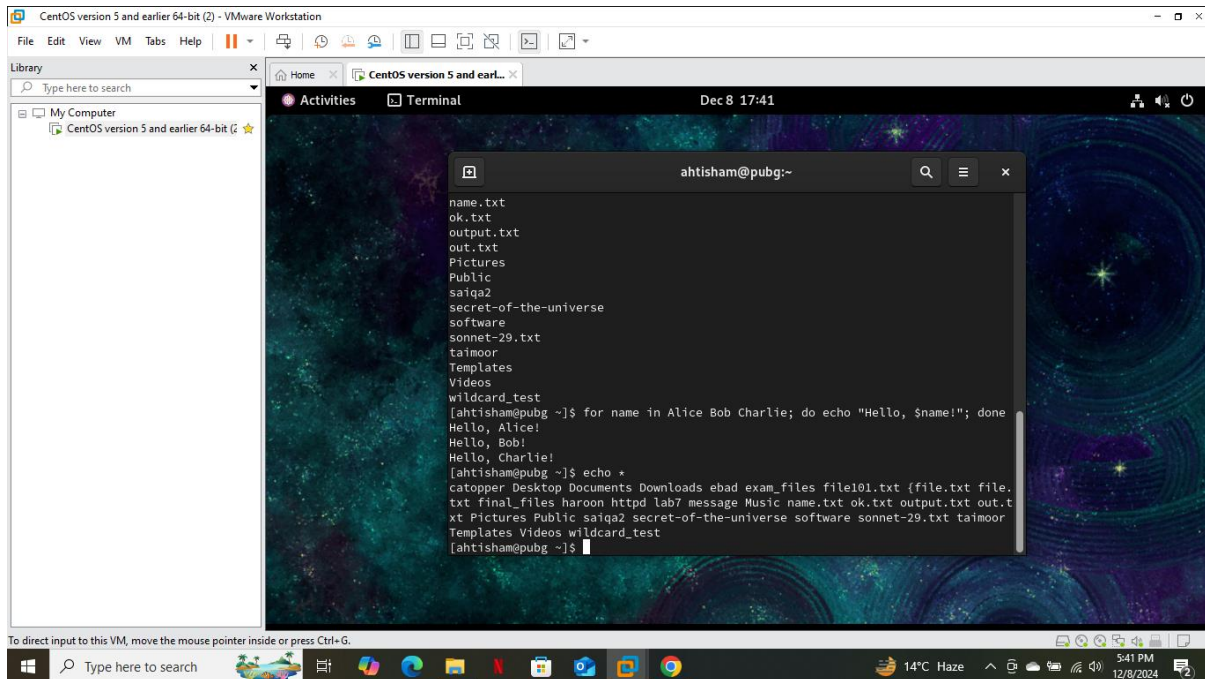


The screenshot shows a CentOS 5 VM window with a terminal open. The terminal displays the following commands and their outputs:

```
[ahtisham@pubg ~]$ ls -a
.                  Desktop      haroon      ok.txt       taimoor
..                 Documents   httpd       output.txt   Templates
.bash_history      Downloads  lab7        out.txt      Videos
.bash_logout       ebad       .lessht     Pictures     .viminfo
.bash_profile      exam_files .local      Public       wildcard_test
.bashrc            file101.txt message     saiqa2
.cache             {file.txt .mozilla   secret-of-the-universe
catoppper          file.txt   Music       software
.config            final_files name.txt    sonnet-29.txt

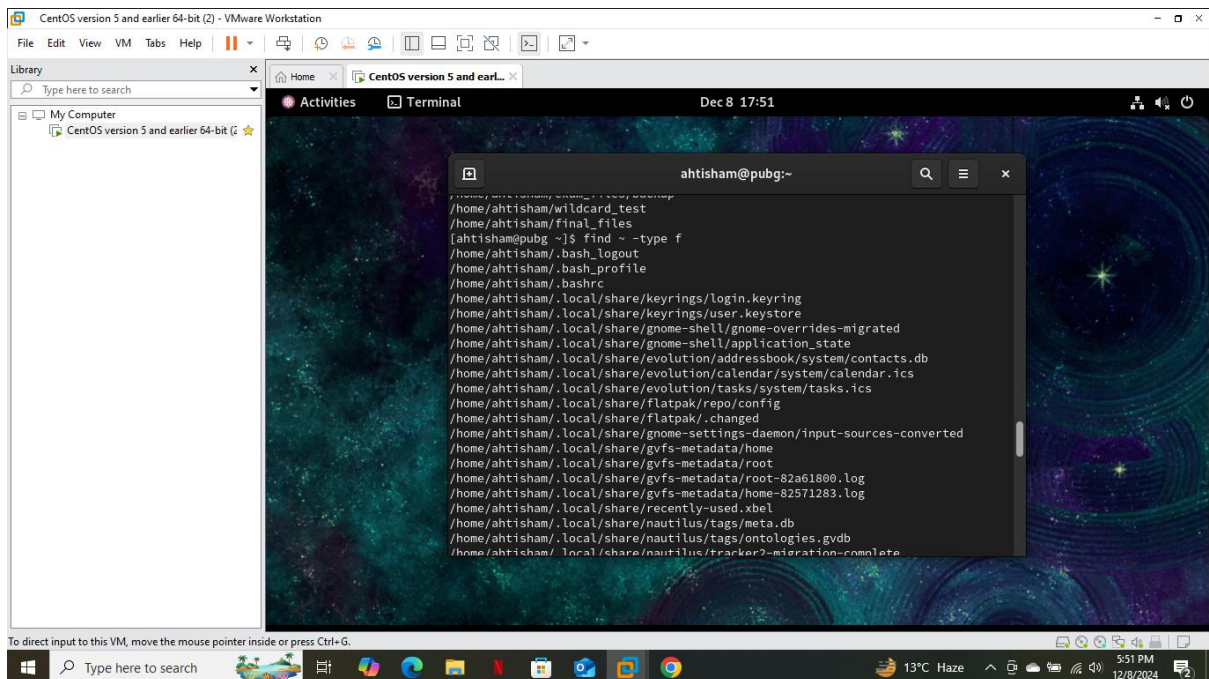
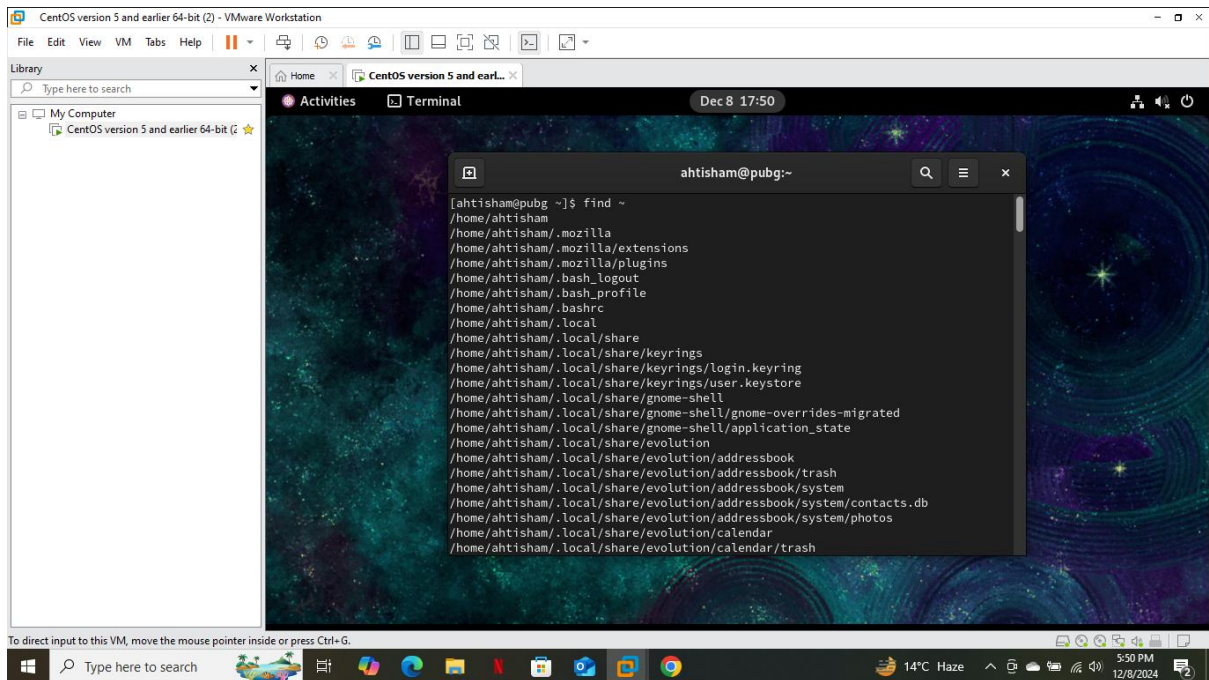
[ahtisham@pubg ~]$ ls -A
.                  Desktop      haroon      name.txt     software
..                 Documents   httpd       ok.txt       sonnet-29.txt
.bash_history      Downloads  lab7        output.txt   taimoor
.bash_logout       ebad       .lessht     out.txt      Templates
.bash_profile      exam_files .local      Pictures     Videos
.bashrc            file101.txt .local      Public       .viminfo
.cache             {file.txt .mozilla   saiqa2       wildcard_test
catoppper          file.txt   Message     secret-of-the-universe
catoppper          Desktop
catoppper          Documents
catoppper          Downloads
```

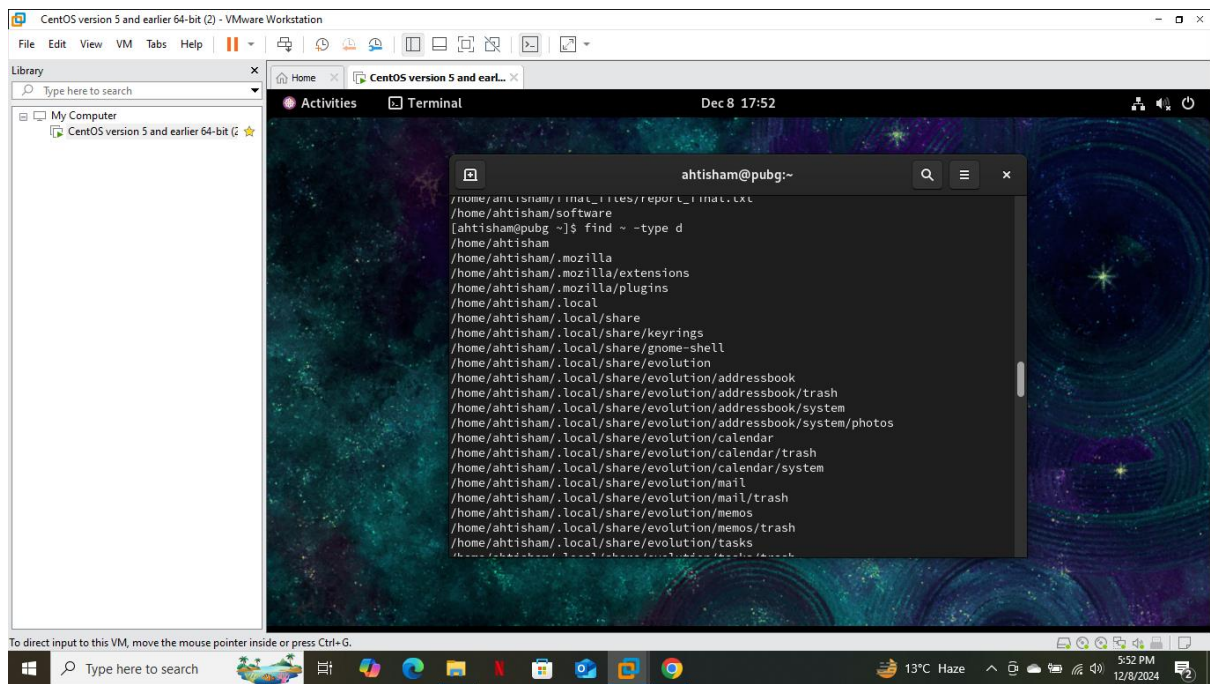
The terminal also shows a for loop command: `for file in *; do echo $file; done`, which prints the names of the files in the current directory.

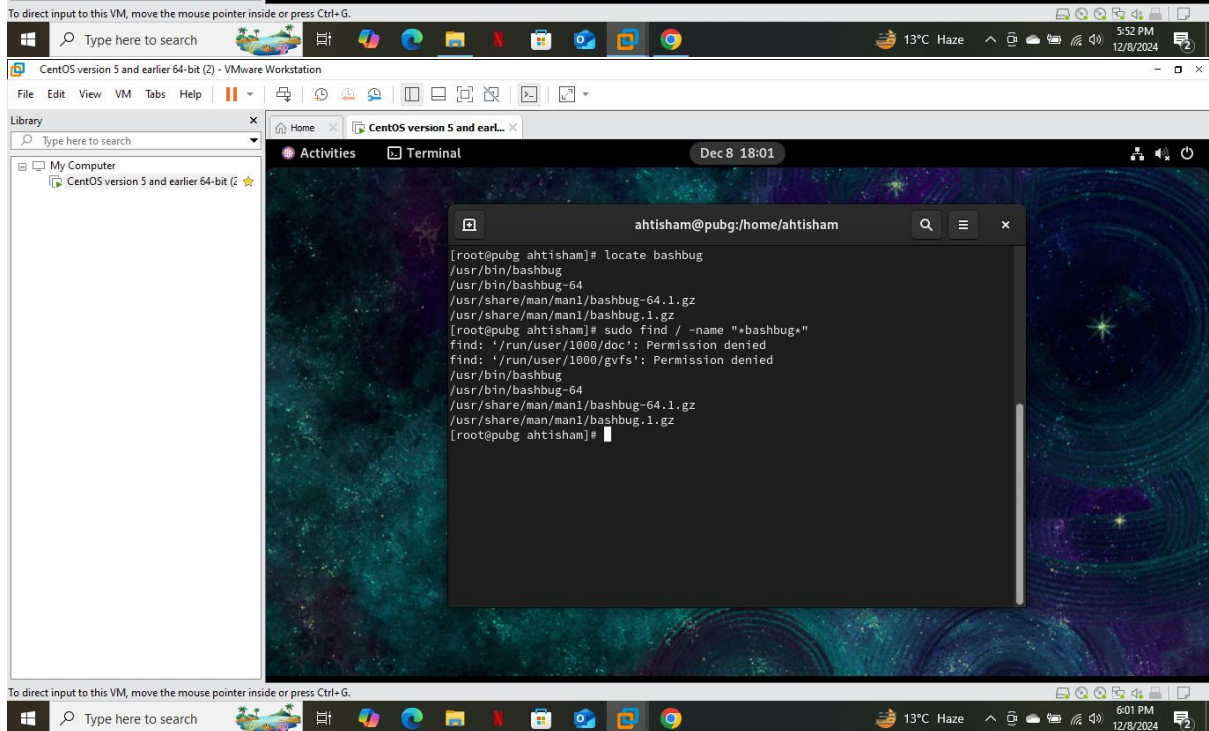
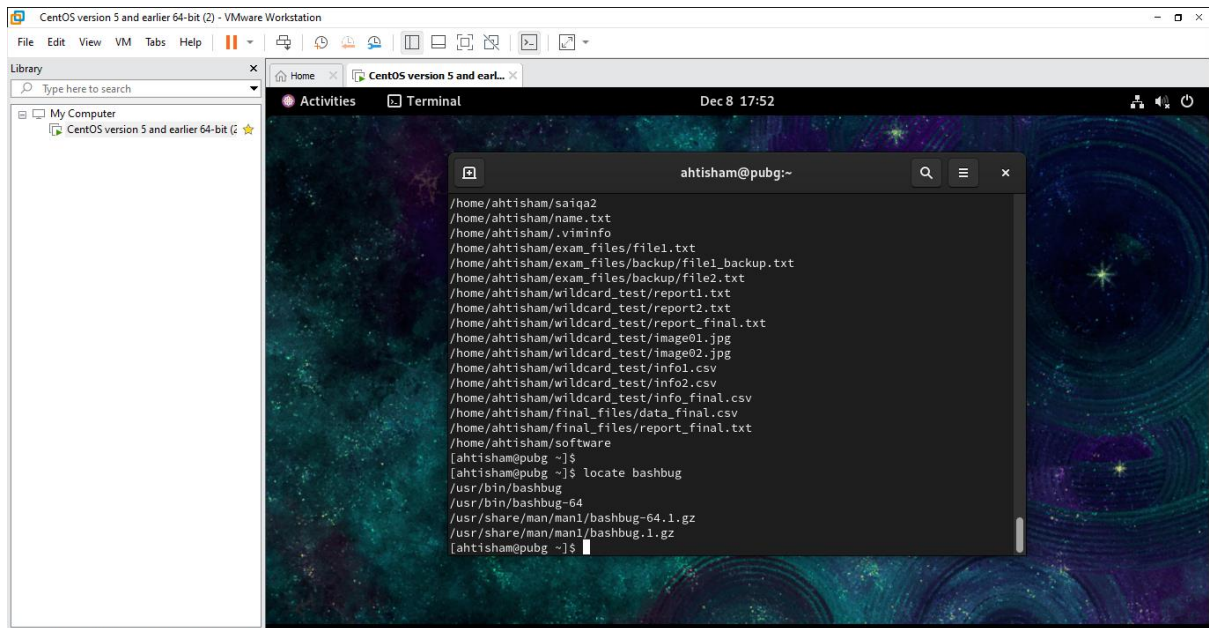


Q3.

- Use the find command to list all the files and directories under your home directory. Try the -type d and -type f criteria to show just files and just directories.
- Use 'locate' to find files whose name contains the string 'bashbug'. Try the same search with find, looking over all files on the system. You'll need to use the * wildcard at the end of the pattern to match files with extensions.







LAB 05.

Q1

- a. Use `cd` to go to your home directory, and create a new directory there called `dog`.
- b. Create another directory within that one called `cat`, and another within that called `mouse`.
- c. Remove all three directories. You can either remove them one at a time, or all at once.
- d. If you can delete directories with `rm -r`, what is the point of using `rmdir` for empty directories?

Answer:

- `rm -r` removes directories and their contents, making it more powerful but also more dangerous if used incorrectly.
 - `rmdir` is safer as it only works on empty directories, preventing accidental deletion of files.
-
- e. Try creating the `dog/cat/mouse` directory structure with a single command.

```
[ahtisham@pubg ~]$ cd
[ahtisham@pubg ~]$ mkdir dog
[ahtisham@pubg ~]$ mkdir dog/cat
[ahtisham@pubg ~]$ mkdir dog/cat/mouse
[ahtisham@pubg ~]$ rm -r dog
[ahtisham@pubg ~]$ mkdir -p dog/cat/mouse
[ahtisham@pubg ~]$ ls
catoppper  desktop  file101.txt  httpd  output.txt  software
desktop   file101.txt  lab7        out.txt  sonnet-29.txt
Documents file.txt    message    Pictures  taimoor
dog        file.txt    music      Public    Templates
Downloads  final_files name.txt   saiqa2   Videos
ebad       haseen    ok.txt     secret-of-the-universe  wildcard_test
[ahtisham@pubg ~]$ cd dog
[ahtisham@pubg dog]$ ls
cat
[ahtisham@pubg dog]$ cd cat
[ahtisham@pubg cat]$ ls
mouse
[ahtisham@pubg cat]$
```

Q2

- Copy the file `/etc/passwd` to your home directory, and then use `cat` to see what's in it.
- Rename it to `users` using the `mv` command.
- Make a directory called `programs` and copy everything from `/bin` into it.
- Delete all the files in the `programs` directory.
- Delete the empty `programs` directory and the `users` file.

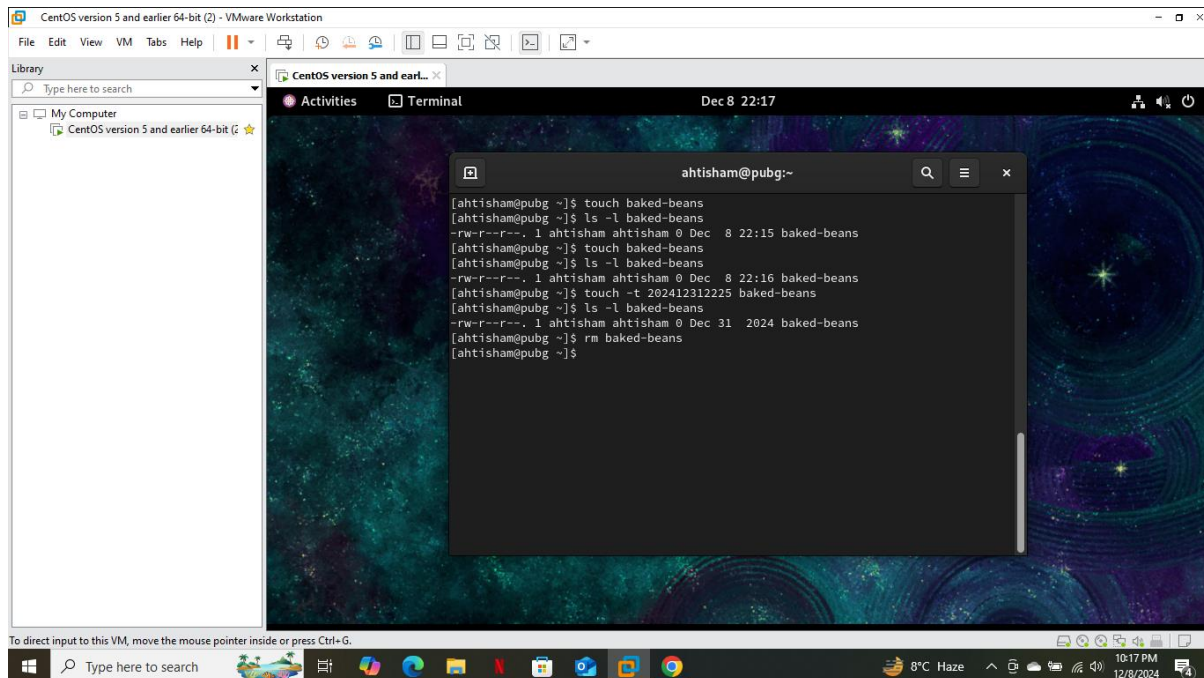
```
Terminal Dec 8 21:56
ahtisham@pubg:/
[ahtisham@pubg ~]$ su
Password:
[root@pubg ahtisham]# cp /etc/passwd /
[root@pubg ahtisham]# cat passwd
cat: passwd: No such file or directory
[root@pubg ahtisham]# cd ..
[root@pubg home]# cd ..
[root@pubg /]# cat passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
systemd-coredump:x:999:997:systemd Core Dumper:/sbin/nologin
dbus:x:81:81:System message bus:/sbin/nologin
```

```
Terminal Dec 8 22:08
ahtisham@pubg:/home/ahtisham
[ahtisham@pubg ~]$ mv /users /passwd
mv: cannot move '/users' to '/passwd': Permission denied
[ahtisham@pubg ~]$ su
Password:
[root@pubg ahtisham]# mv /users /passwd
[root@pubg ahtisham]#
[root@pubg ahtisham]# mv /passwd /users
[root@pubg ahtisham]# mkdir programs
[root@pubg ahtisham]# cp /bin/* programs
[root@pubg ahtisham]# rm -r programs/*
rm: remove regular file 'programs/['?'
rm: remove regular file 'programs/ac'? ^[[A^C
[root@pubg ahtisham]# rm -rf programs/*
[root@pubg ahtisham]# rmdir programs/
[root@pubg ahtisham]# rm /users
rm: remove regular file '/users'? yes
[root@pubg ahtisham]# rm /users
rm: cannot remove '/users': No such file or directory
[root@pubg ahtisham]#
```

Q3

- The touch command can be used to create new empty files. Try that now, picking a name for the new file: `$ touch baked-beans`
- Get details about the file using the ls command: `$ ls -l baked-beans`
- Wait for a minute, and then try the previous two steps again, and see what changes. What happens when we don't specify a time to touch?

d. Try setting the timestamp on the file to a value in the future. e. When you're finished with it, delete the file.



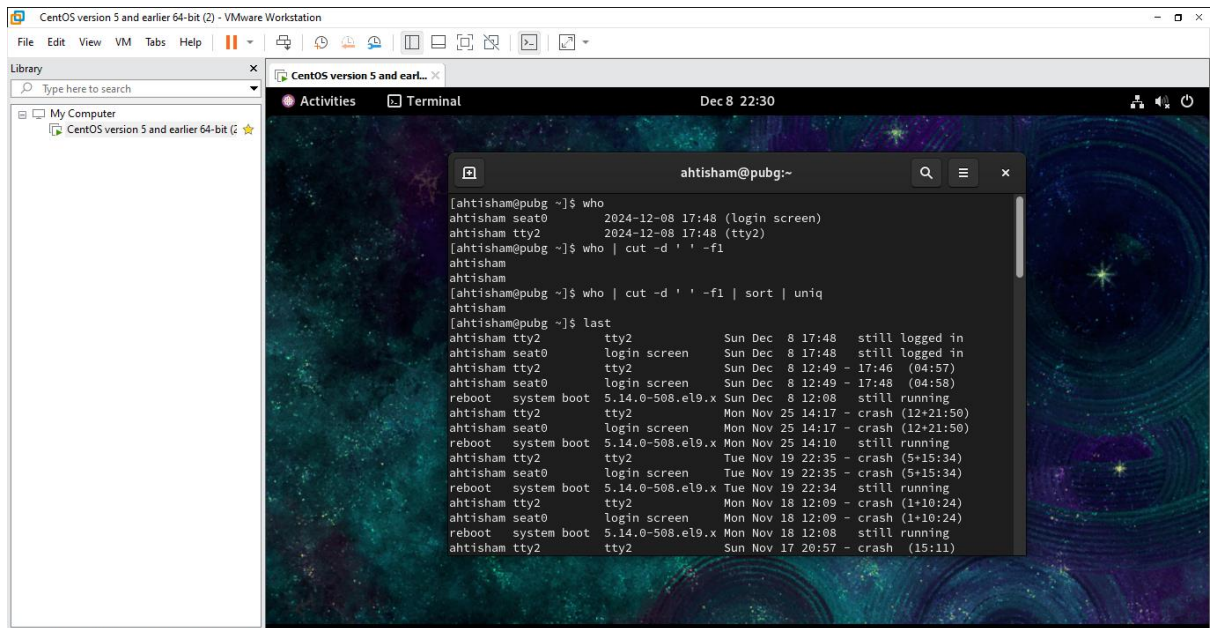
The screenshot shows a VMware Workstation window titled 'CentOS version 5 and earlier 64-bit (2) - VMware Workstation'. Inside the VM, a terminal window is open with the prompt 'ahtisham@pubg:~'. The terminal shows the following commands and output:

```
[ahtisham@pubg ~]$ touch baked-beans
[ahtisham@pubg ~]$ ls -l baked-beans
-rw-r--r--. 1 ahtisham ahtisham 0 Dec  8 22:15 baked-beans
[ahtisham@pubg ~]$ touch baked-beans
[ahtisham@pubg ~]$ ls -l baked-beans
-rw-r--r--. 1 ahtisham ahtisham 0 Dec  8 22:16 baked-beans
[ahtisham@pubg ~]$ touch -t 202412312225 baked-beans
[ahtisham@pubg ~]$ ls -l baked-beans
-rw-r--r--. 1 ahtisham ahtisham 0 Dec 31 2024 baked-beans
[ahtisham@pubg ~]$ rm baked-beans
[ahtisham@pubg ~]$
```

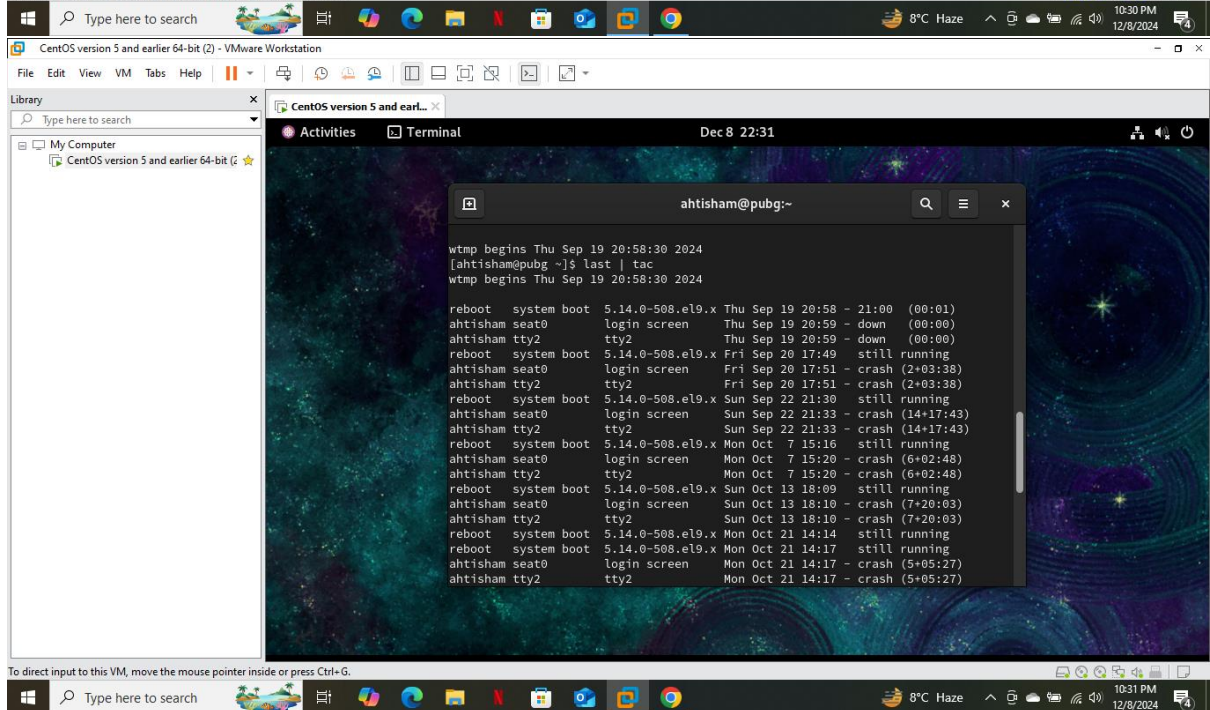
Lab 06.

Q1

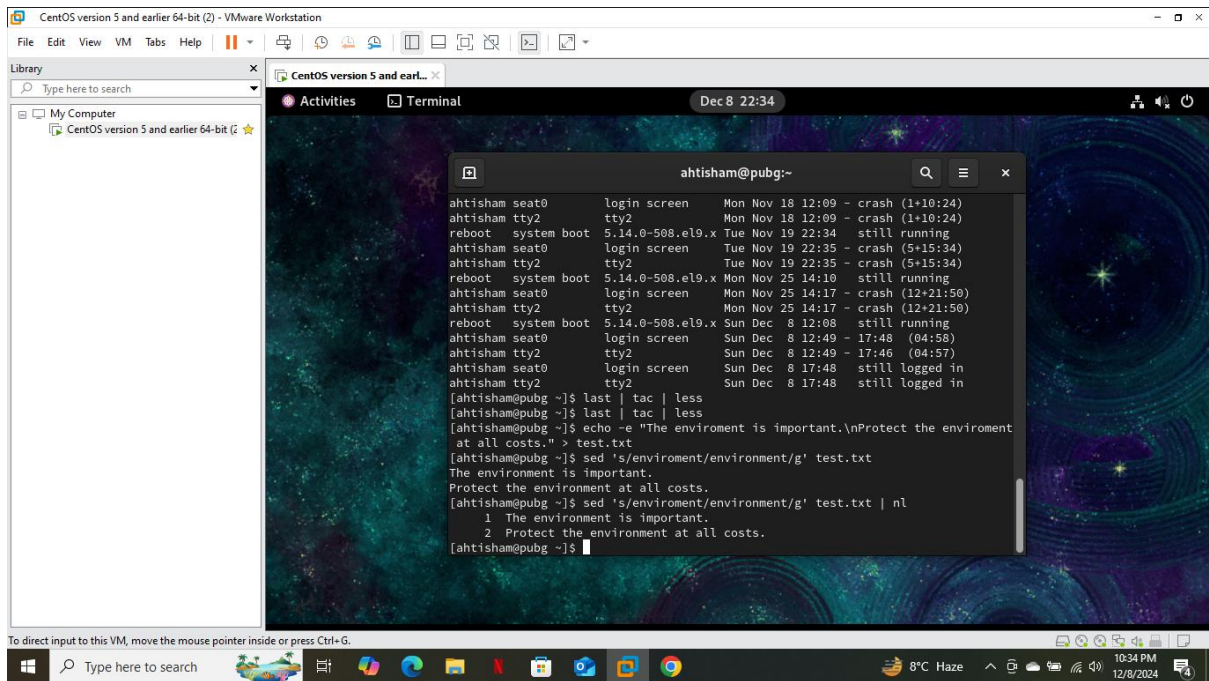
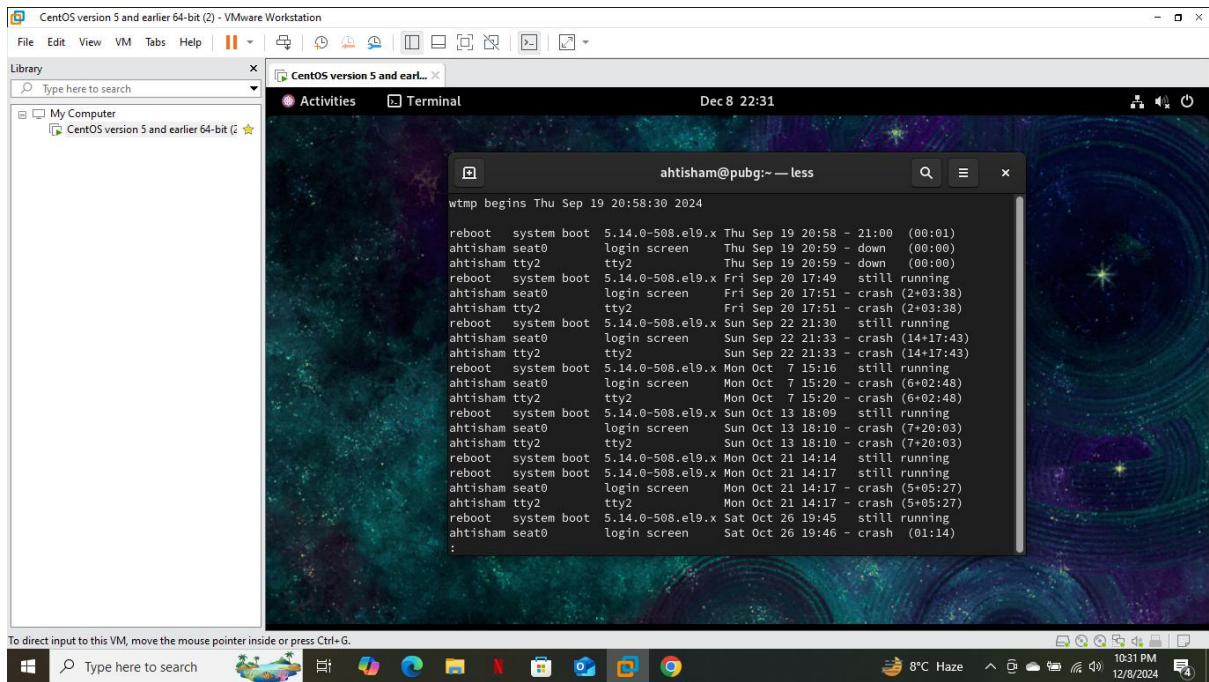
- Type in the example on the cut slide to display a list of users logged in. (Try just who on its own first to see what is happening.)
- Arrange for the list of usernames in who's output to be sorted, and remove any duplicates.
- Try the command last to display a record of login sessions, and then try reversing it with tac. Which is more useful? What if you pipe the output into less?
- Use sed to correct the misspelling 'enviroment' to 'environment'. Use it on a test file, containing a few lines of text, to check it. Does it work if the misspelling occurs more than once on the same line?
- Use nl to number the lines in the output of the previous question.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



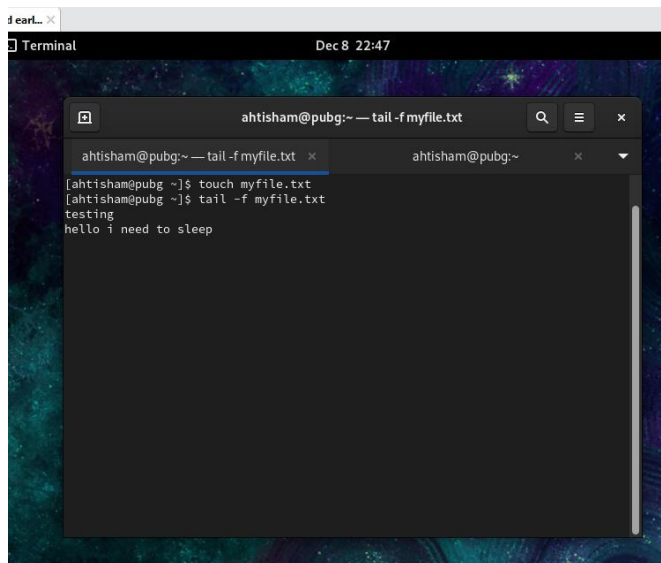
a. Try making an empty file and using `tail -f` to monitor it. Then add lines to it from a different terminal using a command like this: `$ echo "testing" >>filename`

b. Once you have written some lines into your file, use `tr` to display it with all occurrences of the letters A–F changed to the numbers 0–5.

c. Try looking at the binary for the `ls` command (`/bin/ls`) with `less`. You can use the `-f` option to force it to display the file, even though it isn't text.

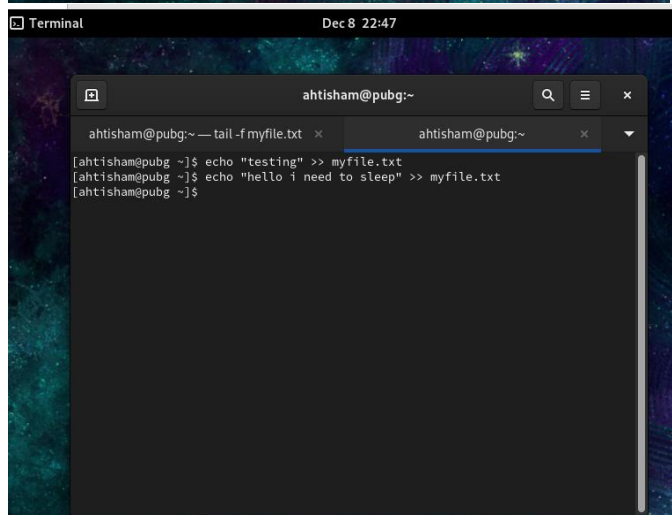
d. Try viewing the same binary with `od`. Try it in its default mode, as well as with the options shown on the slide for outputting in hexadecimal.

od -x /bin/ls



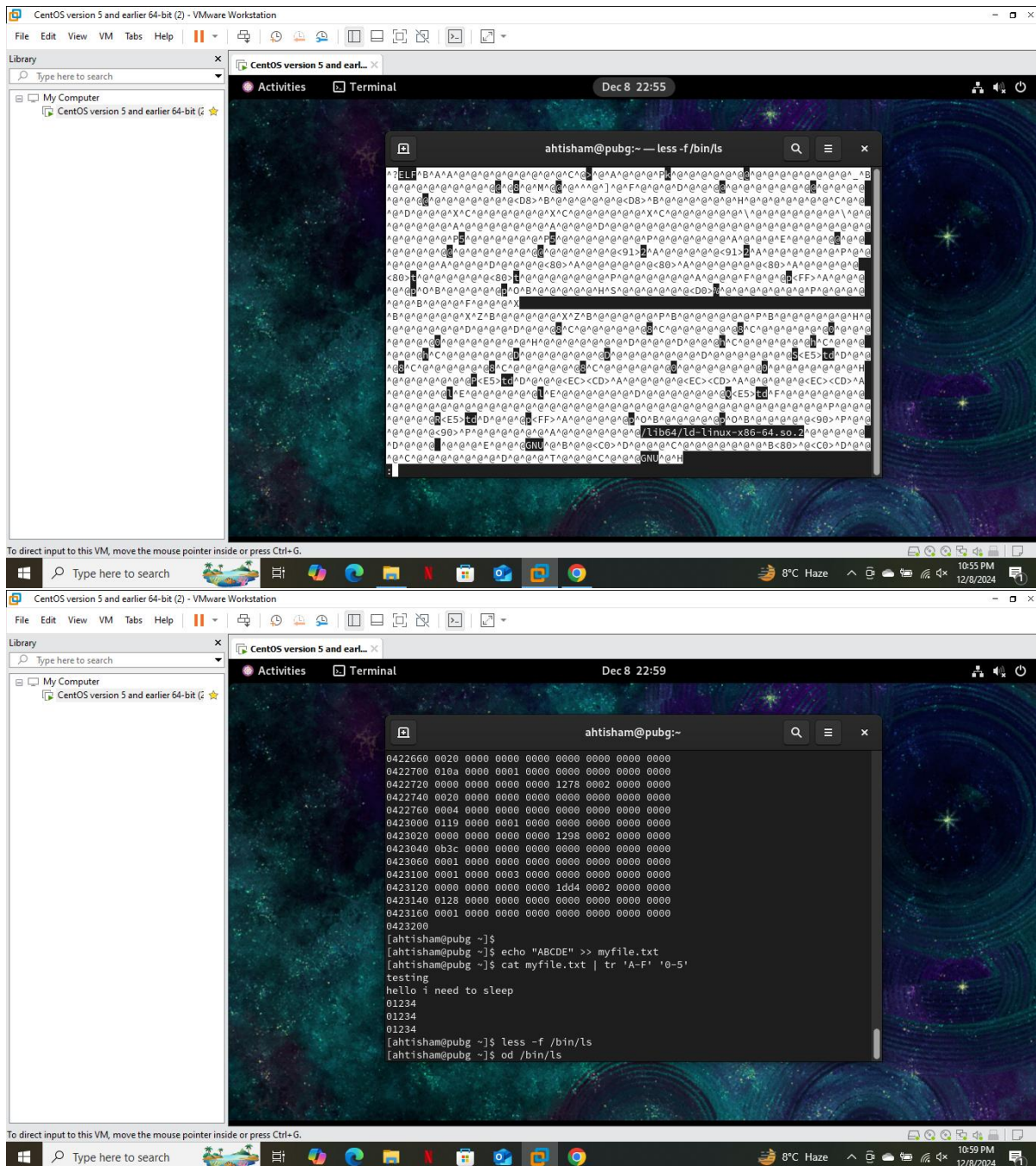
A terminal window titled "Terminal" with a timestamp of "Dec 8 22:47". It shows a sub-window titled "ahtisham@pubg:~ — tail -f myfile.txt". The sub-window contains the following commands and output:

```
ahtisham@pubg:~ — tail -f myfile.txt x ahtisham@pubg:~ x
[ahtisham@pubg ~]$ touch myfile.txt
[ahtisham@pubg ~]$ tail -f myfile.txt
testing
hello i need to sleep
```



A terminal window titled "Terminal" with a timestamp of "Dec 8 22:47". It shows a sub-window titled "ahtisham@pubg:~". The sub-window contains the following commands and output:

```
ahtisham@pubg:~ — tail -f myfile.txt x ahtisham@pubg:~ x
[ahtisham@pubg ~]$ echo "testing" >> myfile.txt
[ahtisham@pubg ~]$ echo "hello i need to sleep" >> myfile.txt
[ahtisham@pubg ~]$
```

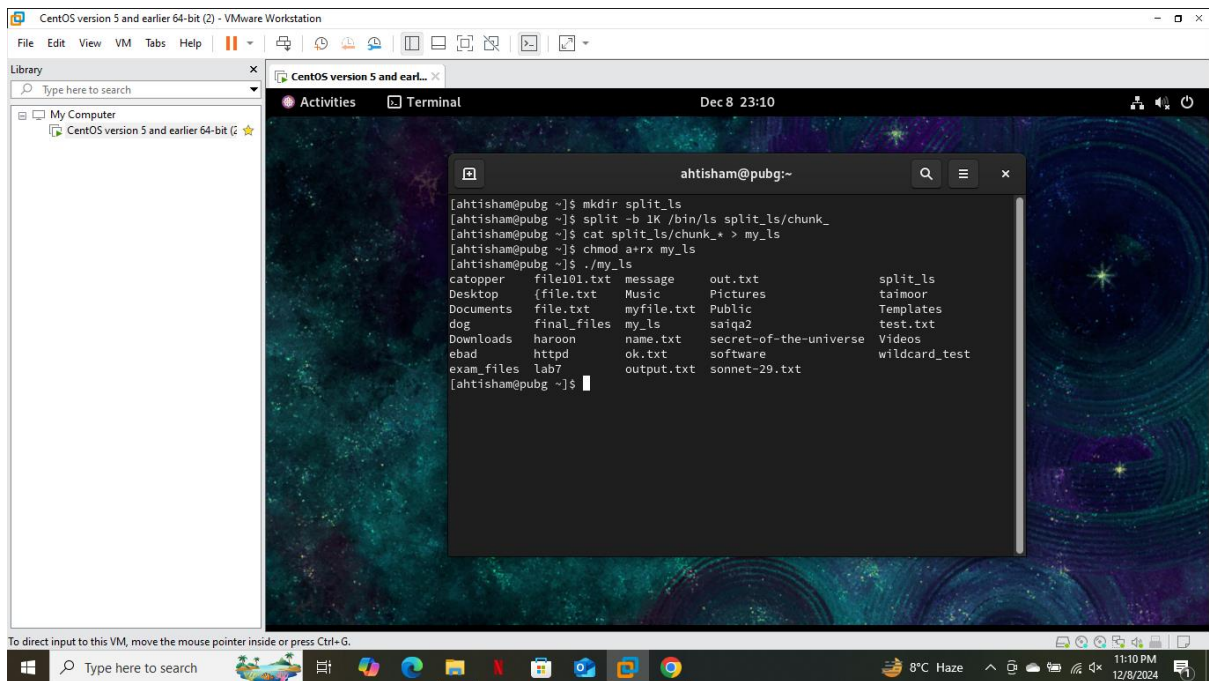


Q3

- Use the split command to split the binary of the ls command into 1Kb chunks. You might want to create a directory especially for the split files, so that it can all be easily deleted later
- Put your split ls command back together again, and run it to make sure it still works. You will have to make sure you are running the new copy of it, for example ./my_ls, and make sure that the program is marked as 'executable'

to run it, with the following command:

```
$ chmod a+rx my_ls
```



```
CentOS version 5 and earlier 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
CentOS version 5 and earlier 64-bit (2)
Activities Terminal Dec 8 23:10
ahtisham@pubg:~
[ahtisham@pubg ~]$ mkdir split_ls
[ahtisham@pubg ~]$ split -b 1K /bin/ls split_ls/chunk_
[ahtisham@pubg ~]$ cat split_ls/chunk_* > my_ls
[ahtisham@pubg ~]$ chmod a+rx my_ls
[ahtisham@pubg ~]$ ./my_ls
catoppe  file101.txt  message  out.txt      split_ls
Desktop {file.txt  Music    Pictures    taimoor
Documents file.txt    myfile.txt Public      Templates
dog      final_files my_ls     saiga2       test.txt
Downloads haroon      name.txt  secret-of-the-universe Videos
ebad     httpd       ok.txt    software     wildcard_test
exam_files lab7       output.txt sonnet-29.txt
[ahtisham@pubg ~]$
```